
Distribution-Calibrated Inference Time Compute for Thinking LLM-as-a-Judge

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Abstract

Thinking Large Language Models (LLMs) used as judges for pairwise preferences remain noisy at the single-sample level, and common aggregation rules (majority vote, soft self-consistency, or instruction-based self-aggregation) are inconsistent when ties are allowed. We study *inference-time compute* (ITC) for evaluators that generate n independent thinking–rating samples per item, and propose a principled, distribution-calibrated aggregation scheme. Our method models three-way preferences with a Bradley–Terry–Davidson formulation on rating counts, leveraging both *polarity* (margin among non-ties) and *decisiveness* (non-tie rate) to distinguish narrow margins from strong consensus. Across various evaluation benchmarks, our approach consistently reduces MAE and increases pairwise accuracy versus standard baselines, and when evaluated against human-consensus meta-labels, matches or exceeds individual human raters. These results show that carefully allocating ITC and aggregating with distribution-aware methods turns noisy individual model judgments into reliable ratings for evaluation.

1. Introduction

Thinking large language models (LLMs) are increasingly employed as automated judges for evaluating the output of other generative systems, a paradigm known as “Thinking-LLM-as-a-Judge” (Saha et al., 2025). This approach offers a scalable and cost-effective alternative to human evaluation, which is often slow and expensive. To mitigate the inherent stochasticity and noise of single-pass judgments, a common strategy is to leverage inference-time compute

(ITC) (Snell et al., 2025) by generating multiple independent reasoning and rating samples for each item being evaluated. However, the reliability of the final judgment hinges critically on how these multiple outputs are aggregated.

Current aggregation methods, such as majority voting (Self-Consistency) (Wang et al., 2023) or heuristics based on model confidence scores or LLM generated aggregators, are often brittle and statistically suboptimal. These approaches are particularly fragile in the presence of ties. For instance, a simple majority vote cannot distinguish between a narrow 5-to-4 decision and a decisive 9-to-0 consensus, discarding valuable information about the strength of evidence contained within the full distribution of votes. This insensitivity to evidential strength leads to less reliable and robust evaluations.

In this work, we argue that the aggregation step is not an afterthought but a critical component for effectively utilizing ITC. We propose a principled, Distribution-Calibrated Aggregation scheme that moves beyond simple vote-counting. Intuitively, instead of applying a rigid majority rule, we calibrate a probabilistic model on a small set of labeled examples so it can adapt to the LLM judge’s specific voting behavior—such as its unique propensity to declare ties or push for decisive wins. To achieve this, our method operates directly on the full counts of positive, negative, and tie votes, preserving the full signal in the sample distribution. Specifically, we model these three-way preference outcomes using a Bradley-Terry-Davidson (Davidson, 1970) formulation, which explicitly parametrizes both the preference margin and the tie propensity. At inference, we apply a decision rule designed to explicitly minimize our target evaluation metric, Mean Absolute Error (MAE). By directly aligning the aggregation with this ordinal-aware metric—penalizing complete preference reversals more heavily than minor tie disagreements—we avoid loss-metric mismatch and yield more accurate judgments. Conceptually, this calibration step modifies the decision boundary compared to simple majority voting, as demonstrated in Figure 1.

We conduct extensive experiments on a diverse set of benchmarks, including MT (Machine Translation) evaluation (WMT23) (Song et al., 2025) and reward model as-

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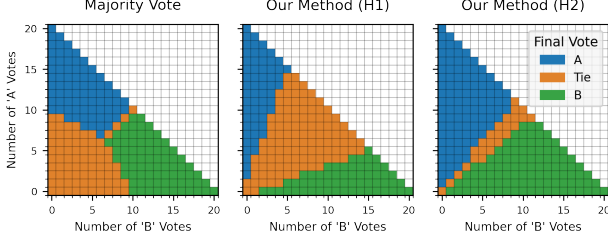


Figure 1. Behavior of Different Aggregation Methods with 20 Votes. The number of ‘Tie’ votes is computed as $20 - (\# \text{ of A votes}) - (\# \text{ of B votes})$. Our proposed method’s behavior is shown using two different calibrated parameter sets: H1 models a high-tie propensity regime, while H2 models a tie-averse regime.

assessment (Reward Bench 2) (Malik et al., 2025). Our results show that our distribution-calibrated approach considerably outperforms a suite of strong self-consistency baselines. By carefully modeling the entire vote distribution, our method turns noisy individual model judgments into more reliable ratings, matching or exceeding the performance of individual human raters when evaluated against a human-consensus gold standard.

Contributions: Our main contributions are threefold: (1) We show that the existing aggregation methods for inference time compute for LLM judges are suboptimal and that a carefully designed aggregation approach is critical. (2) We propose an Expected Risk Minimization (ERM)-based Bradley–Terry–Davidson aggregation fit on a small calibration set, and show that it consistently outperforms existing aggregation methods across different tasks in both reward benchmarks and MT. (3) For MT in particular, we adopt a consensus-based meta-evaluation to form higher-fidelity ground truths where labels are noisy, enabling fair comparison to human raters and revealing regimes where LLM judges approach “super-human” evaluation quality.

Conflict of Interest Disclosure. All authors are employed by Google-affiliated organizations. Google develops the Gemini model family, and this paper evaluates gemini-2.5-flash (Comanici et al., 2025) as one of the LLM judges. The proposed aggregation method is model-agnostic, and our experiments also include non-Google and open-source models.

2. Related Work

LLM-as-a-Judge: Recently, Large Language Models (LLMs) have achieved remarkable success when deployed as “judges” (Zheng et al., 2023) to evaluate generated text, offering a scalable alternative to traditional metrics (Gu et al., 2025). This paradigm has demonstrated high correlation with human judgments across diverse domains. Approaches vary: some prompt general-purpose LLMs

directly (e.g., G-Eval (Liu et al., 2023); JudgeLM (Zhu et al., 2025)), while others fine-tune specialized models optimized for evaluation tasks (e.g., Prometheus (Kim et al., 2024); Auto-J (Li et al., 2024)). While powerful, these LLM-based approaches face significant challenges, including sensitivity to prompt design (Gu et al., 2025) and inherent biases, such as positional bias (favoring a specific candidate order) or verbosity bias (preferring longer outputs) (Wang et al., 2024b). Moreover, LLM judges exhibit variability in their decision-making, with some models being more aggressive than others in breaking subtle distinctions or ties (Zheng et al., 2023). Our work focuses on mitigating this noise and improving the reliability of judgments through a principled aggregation.

Thinking in Language Models for Evaluation. The reliability of LLM judgments is often enhanced when the model is prompted to generate intermediate reasoning steps before emitting a final verdict, a technique popularized by Chain-of-Thought (CoT) prompting (Wei et al., 2022). In the context of evaluation, this *thinking* process allows the model to articulate the criteria for judgment and justify its decision, leading to the *Thinking-LLM-as-a-Judge* paradigm (Saha et al., 2025). This explicit reasoning not only improves the accuracy of the judgments (Zhang et al., 2025) but also increases their interpretability. Our work leverages the generation of these independent thinking traces and investigates how to best aggregate the resulting rating samples.

Inference Time Compute and Sample Aggregation: Multiple strategies have been proposed that leverage inference time compute in reward modeling and evaluation (Liu et al., 2025). When multiple samples are generated, an aggregation strategy is required. Self-Consistency (SC) (Wang et al., 2023) aggregates multiple outputs using majority voting. Several variants incorporate confidence signals. Soft Self-consistency (Soft-SC) (Wang et al., 2024a) picks the minimum, mean, or product of confidence scores of items in each category. Confidence-Informed Self-Consistency (CI-SC) (Taubenfeld et al., 2025) computes a weighted majority vote based on confidence scores, which are computed as either the length-normalized probability of the sequence or via prompting an LLM. Alternatively, some methods leverage the LLM itself for aggregation. Generative Self-Aggregation (GSA) (Li et al., 2025) asks the LLM to synthesize a new response based on the context of multiple samples. Universal Self-Consistency (USC) (Chen et al., 2023) leverages the LLM to select the most consistent answer among multiple candidates. Furthermore, Singhi et al. (2025) and Zhang et al. (2025) showed that one can improve the performance of reasoning-based generative verifiers via test-time compute, particularly via majority voting. Concurrent to our work, AtC (Xie et al., 2026) addresses human-centered assessment by calibrating continuous model scores to an externally derived human

Table 1. Allowing a ‘Tie’ Option Reduces Positional Bias. The table compares preferences in a forced-choice setting against one where a ‘tie’ is allowed. The bias is computed as $(\#First - \#Second) / (\#First + \#Second)$

Model	Forced-Choice (No Tie)			Tie Allowed			
	First	Second	Bias	First	Second	Tie	Bias
gemini-2.5-flash	385	287	14.5%	220	199	253	3.1%
qwen3-next-80b	309	363	-8.0%	322	318	32	0.6%

Table 2. Ties Rates for Different Models and Prompts (in %) with 95% confidence intervals.

Model	prompt_1	prompt_2	prompt_3	Model Avg
gpt-oss-120b	19.3% ($\pm 4.2\%$)	24.4% ($\pm 4.6\%$)	21.6% ($\pm 4.4\%$)	21.8% ($\pm 2.5\%$)
gemini-2.5-flash	12.4% ($\pm 3.5\%$)	21.3% ($\pm 4.4\%$)	37.6% ($\pm 5.2\%$)	23.8% ($\pm 2.6\%$)
deepseek-v3.1	28.9% ($\pm 4.8\%$)	29.6% ($\pm 4.9\%$)	32.6% ($\pm 5.0\%$)	30.4% ($\pm 2.8\%$)
Prompt Avg	20.2% ($\pm 2.5\%$)	25.1% ($\pm 2.7\%$)	30.6% ($\pm 2.8\%$)	Overall: 25.3% ($\pm 1.5\%$)

consensus ranking via isotonic regression. By contrast, our method calibrates a probabilistic aggregator directly for three-way LLM preference judgments with ties.

Generator Refinement and Verification: A different line of work refines the generation process. Methods like Mirror-Consistency (Huang et al., 2024), Self-Contrast (Zhang et al., 2024), and Step-Back Prompting (Zheng et al., 2024) utilize iterative reflection or diverse perspectives to produce higher-quality samples, while Self-Check (Miao et al., 2024) employs step-wise verification to filter errors. Unlike these approaches, which focus on enhancing the generator (often incurring sequential computational costs), our work focuses on the aggregator: we accept the noisy distribution of parallel samples and apply a calibration layer to robustly estimate the ground truth.

3. Motivation: The Tie Dilemma

A critical choice when designing an LLM-as-a-Judge for pairwise comparisons (Zheng et al., 2023) is whether to allow the judge to declare a tie or force it to pick a preference. In this section, we first show that forcing the model to break ties might induce LLM biases. We then show that the tie decisions are highly sensitive to the judge parameters, requiring a more robust aggregation method to mitigate.

Ties are important to reduce LLM biases: LLM-as-a-Judge exhibit multiple types of systematic biases (Ye et al., 2025a). A well-known issue is positional bias (Shi et al., 2025), where the model’s preference can be affected by the order in which responses are presented.

To quantify this, we evaluated qwen3-next-80b (Qwen Team, 2025), gpt-oss-120b (OpenAI, 2025), deepseek-v3.1 (DeepSeek-AI, 2024), and gemini-2.5-flash (Comanici et al., 2025) on a subset of 336 human-tie pairs from the WMT23 ZH $\rightarrow$ EN dataset (detailed in Section 5). By eval-

uating each pair in both orders, we measured positional bias in a forced-choice setting. Table 1 presents the two models exhibiting notable bias: gemini-2.5-flash favors the first position by 14.58%, while qwen3-next-80b favors the second by 8.04%. The remaining models (gpt-oss-120b and deepseek-v3.1) showed negligible bias ($< 1\%$) and are omitted.

The right side of Table 1 shows the results from the same experiment but with the prompt updated to allow ties. The introduction of this third choice dramatically reduces positional bias for both models. This demonstrates that including a tie option is not just a feature for capturing equivalence, but might be a critical mechanism for debiasing the evaluation process itself.

Tie decisions are not stable: Our work is motivated by the fact that in three-way preference tasks, the vote distribution of an LLM-as-a-judge is highly sensitive to variations in the evaluation setup.

In this section, we demonstrate empirically two major sources of variability in ratings - (1) the LLM queried and (2) the prompt template used to get the ratings. We conduct an experiment where we generated three slight variations of an evaluation prompt as shown in Appendix A. We then use each of these prompts to judge the same dataset from the previous section. As shown in Table 2, the results reveal a significant variance in the rate of ties across prompts and LLMs. To quantify data-sampling variance, we report 95% confidence intervals. For instance, using the gemini-2.5-flash model, the percentage of ‘‘Ties’’ votes fluctuates dramatically, ranging from a high of 37.6% ($\pm 5.2\%$) with prompt_3 to a low of 12.4% ($\pm 3.5\%$) with prompt_1. Because these confidence bounds do not overlap, this confirms the prompt-induced shifts are highly statistically significant. We also observe that deepseek-v3.1 produces an average tie rate of 30.4% across all prompts, which is sig-

nificantly higher than gpt-oss-120b’s average of 21.8%.

This instability is a critical flaw for methods that do not calibrate for such variations since a simple change in prompt wording can fundamentally alter the tie likelihood. This underscores the need for a robust distribution-calibrated aggregation method, which can explicitly model and adapt to these shifts. Other works have investigated calibration via finetuning the model (Park et al., 2024) (Ye et al., 2025b); in this work we focus on mitigation strategies at inference time.

4. Distribution-Calibrated Inference-Time Sample Aggregation

Setting. Given a prompt x and a pair of responses (A, B) , our autorater queries a Thinking LLM n times to obtain independent reasoning–rating tuples $\{(z_j, r_j)\}_{j=1}^n$, where z_j is a thinking trace and $r_j \in \{-1, 0, +1\}$ is a discrete vote ($+1$: $A \succ B$, -1 : $B \succ A$, 0 : $A \sim B$). Empirically, once a thinking trace z_j is produced, the conditional distribution $p(r_j \mid z_j, \cdot)$ is sharply peaked (Wang et al., 2025). In addition, we do not see a high variation in the normalized probability of the thinking traces. We therefore find that log-likelihood reweighting adds little signal in practice. Instead, we operate directly on the vote counts, which preserve the strength of evidence in the sample distribution. Let $c^+ = |\{j : r_j = +1\}|$, $c^- = |\{j : r_j = -1\}|$, $c^0 = |\{j : r_j = 0\}|$, $n = c^+ + c^- + c^0$, and equivalently $\mathbf{n} = (c^+, c^0, c^-)$. While majority vote (the mode of $\mathbf{n}$) is common, it is statistically suboptimal: it is highly sensitive to sampling noise and ignores evidential strength (e.g., it cannot distinguish 5-to-4 from 9-to-0). We instead aggregate via a parametric model that consumes the full count distribution and is aligned to our evaluation metric.

Evaluation Metric. For a dataset of N items, let $y_i^* \in \{-1, 0, +1\}$ denote the ground truth and $\hat{y}_i$ the aggregator’s decision. We evaluate with mean absolute error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N \ell(\hat{y}_i, y_i^*), \quad \ell(a, b) = |a - b|. \quad (1)$$

This ordinal-aware metric is well-suited for the ordered label set $\{-1, 0, +1\}$. Unlike standard accuracy, which treats all misclassifications uniformly, MAE scales penalties by severity: it penalizes complete preference reversals (error of magnitude 2) more heavily than tie-related disagreements (error of magnitude 1), thereby preserving the semantic hierarchy of the preference scale.

Count-derived features from votes. We extract two smoothed features from $\mathbf{n}$:

$$s = \frac{1}{2} \log \frac{c^+ + \alpha}{c^- + \alpha}, \quad (2)$$

with small $\alpha > 0$ (we use $\alpha=1$), capturing the decisive margin; and a tie-evidence feature

$$t = \log \frac{c^0 + \kappa}{n + \kappa} \leq 0, \quad (3)$$

with $\kappa > 0$ (we use $\kappa=1$), which increases (toward 0) as ties appear more frequently.

A Davidson-style model with ties. We adopt a multinomial logit model inspired by the Bradley–Terry–Davidson framework for ternary outcomes. For an item with a latent margin $u \in \mathbb{R}$ and a tie logit $\eta \in \mathbb{R}$,

$$p(+1) = \frac{e^u}{Z}, \quad p(-1) = \frac{e^{-u}}{Z}, \quad p(0) = \frac{e^\eta}{Z}, \quad (4)$$

where $Z = e^u + e^{-u} + e^\eta$. We link features to scores linearly and *jointly* model both decisive margin and tie propensity:

$$u = \beta s \quad \text{and} \quad \eta = \eta_0 + \gamma t, \quad (5)$$

with parameters $\theta = (\beta, \eta_0, \gamma)$. This single specification allows a global tie baseline via η_0 and item-specific modulation via t with slope γ .

MAE-aligned decision rule. Given θ and an input (s, t) , we compute probabilities via Equation 4–Equation 5. The Bayes-optimal action under MAE is the label $y \in \{-1, 0, +1\}$ that minimizes the expected risk:

$$\begin{aligned} \mathcal{R}(-1) &= p(0) + 2p(+1), \\ \mathcal{R}(0) &= p(+1) + p(-1), \\ \mathcal{R}(+1) &= 2p(-1) + p(0). \end{aligned} \quad (6)$$

The optimal decision is therefore given by:

$$\hat{y} = \arg \min_{y \in \{-1, 0, +1\}} \mathcal{R}(y). \quad (7)$$

Parameter fitting via The Discrete Ranked Probability Score. A direct approach is to minimize empirical MAE on a held-out calibration set $\mathcal{C}$:

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \ell(\hat{y}_i(\theta), y_i^*), \quad (8)$$

where $\hat{y}_i$ is obtained by computing (u_i, η_i) from (s_i, t_i) , the Davidson probabilities (Equation 4), and then the MAE Bayes action (Equation 7). However, Equation 8 is ill-suited for standard gradient-based methods as predictions change only when a decision boundary is crossed.

To address this problem, we decouple the model fitting from the decision rule. We fit the probabilistic model by minimizing the *Discrete Ranked Probability Score* (DRPS),

Algorithm 1 Aggregation with calibrated Davidson model

Require: Calibration set $\mathcal{C}$, source query x , response pair (A, B) , sampling budget n , smoothing factors (α, κ) .

Calibrate parameters (offline, once):

- 1: For each $i \in \mathcal{C}$, tally (c_i^+, c_i^-, c_i^0) and compute $s_i = \frac{1}{2} \log \frac{c_i^+ + \alpha}{c_i^- + \alpha}$ and $t_i = \log \frac{c_i^0 + \kappa}{n_i + \kappa}$ (Eqs. 2–3).
- 2: Fit $\hat{\theta} = (\hat{\beta}, \hat{\eta}_0, \hat{\gamma})$ by minimizing empirical DRPS Equation 12 with L-BFGS-B.
- Aggregate a new response pair (online):**
- 3: Query LLM n times to obtain votes $\{r_j\}_{j=1}^n \subset \{-1, 0, +1\}$; tally (c^+, c^-, c^0) ; compute $s = \frac{1}{2} \log \frac{c^+ + \alpha}{c^- + \alpha}$ and $t = \log \frac{c^0 + \kappa}{n + \kappa}$.
- 4: Form $(u, \eta) = (\hat{\beta}s, \hat{\eta}_0 + \hat{\gamma}t)$ and compute $p(-1), p(0), p(+1)$ via Equation 4.
- 5: Compute risks $\mathcal{R}(-1), \mathcal{R}(0), \mathcal{R}(+1)$ via Equation 6.
- 6: **Output** $\hat{y}$ via the Bayes action Equation 7.

a strictly proper scoring rule designed for ordinal outcomes (Gneiting & Raftery, 2007). Let the ordered label set be $\{-1, 0, +1\}$ and define the cumulative probabilities:

$$\begin{aligned} F_{-1} &\equiv \Pr(Y \leq -1) = p(-1), \\ F_0 &\equiv \Pr(Y \leq 0) = p(-1) + p(0). \end{aligned} \quad (9)$$

For an observation y^* , define the corresponding cumulative indicators:

$$\begin{aligned} H_{-1}(y^*) &= \mathbb{1}\{y^* \leq -1\} \\ H_0(y^*) &= \mathbb{1}\{y^* \leq 0\}. \end{aligned} \quad (10)$$

where $\mathbb{1}$ denotes the indicator function. The per-item DRPS is the squared CDF discrepancy:

$$\begin{aligned} \text{DRPS}(p(\cdot | s, t), y^*) &= (F_{-1} - H_{-1}(y^*))^2 \\ &\quad + (F_0 - H_0(y^*))^2. \end{aligned} \quad (11)$$

We then estimate the parameters θ via empirical risk minimization on the calibration set $\mathcal{C}$:

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \text{DRPS}(p_{\theta}(\cdot | s_i, t_i), y_i^*). \quad (12)$$

This approach is preferable to direct MAE minimization for three reasons: (i) **Fisher Consistency.** As a *strictly proper scoring rule* for ordinal outcomes, DRPS is uniquely minimized by the true data-generating distribution (Gneiting & Raftery, 2007). This guarantees Fisher consistency—recovery of the true parameters θ in the population limit. (ii) **Alignment with MAE Decision Rule.** Our final decision action is the MAE Bayes rule in Equation 7, which depends on well-calibrated class probabilities. While MAE is an ordinal-aware metric for point estimates, the DRPS

is its natural generalization to probabilistic forecasts. Minimizing DRPS produces calibrated, ordinal-aware probabilities, ensuring that the downstream Bayes action equation 7 is asymptotically risk-optimal for the MAE metric. (iii) **Superior Optimization Landscape.** Unlike the non-smooth ERM–MAE objective Equation 8, the DRPS objective in Equation 12 is differentiable with respect to θ . This enables efficient estimation using quasi-Newton methods (e.g., L-BFGS-B) under simple box constraints (Nocedal & Wright, 2006).

Hence, we fit the model by minimizing the empirical DRPS on a calibration set and apply the MAE Bayes decision rule at inference time. This two-stage procedure is summarized in Algorithm 1.

5. Experiments

Baselines: In our experiments, we consider the following baselines:

1. Greedy decoding (GD): draws $n = 2$ samples with reversed order and a temperature of zero.
2. Few Shot (FS): draws $n = 2$ samples (with temperature of zero) with reversed order with the labeled calibration set provided in the prompt as in-context examples.
3. Self-Consistency (SC) (Wang et al., 2023): aggregates multiple outputs using majority voting.
4. Soft Self-Consistency (Soft-SC) (Wang et al., 2024a): picks the minimum, mean, or product of confidence scores within each category.
5. Confidence-Informed Self-Consistency (CI-SC) (Taubenfeld et al., 2025): computes a weighted majority vote based on confidence scores; here we use the length-normalized probability of the sequence ($\in [0, 1]$). Alternatively, one could prompt an LLM for the confidence score (Kadavath et al., 2022), but in our experiments the LLM was almost always highly confident.
6. Generative Self-Aggregation (GSA) (Li et al., 2025): asks the LLM to synthesize a new response based on the context of multiple samples.
7. Universal Self-Consistency (USC) (Chen et al., 2023): leverages the LLM to select the most consistent answer among multiple candidates.

In both GD and FS, we aggregate the two responses using a rounded median, where a pair of $(0, 1)$ is mapped to 1. Empirically, this choice leads to better results in both cases. In other baselines, to overcome the positional bias, we draw $\frac{n}{2}$ samples in an A-then-B response order and the remaining $\frac{n}{2}$ samples via a B-then-A order. We then aggregate the entire n samples. In our experiments (except for the GD and FS baselines), we use temperature sampling with $T = 0.5$ to generate the candidates. As detailed in our tem-

perature ablation (Appendix G), maintaining this diversity of reasoning paths is critical; lowering the temperature toward $T = 0$ collapses the samples to the mode, stripping the BTD aggregator of the distributional signal it needs to properly calibrate ties. For LLM aggregation methods, we use greedy decoding in the aggregation stage.

Thinking Models We use the following Thinking LLMs: gemini-2.5-flash (Comanici et al., 2025), qwen3-next-80b (Qwen Team, 2025), gpt-oss-120b (OpenAI, 2025). All the LLM calls in this paper are done through Thinking LLMs with thinking enabled.

Benchmarks We consider two machine translation tasks (Song et al., 2025) and six tasks from the Reward Bench 2 benchmark (Malik et al., 2025). See Appendix A for the prompts.

We use the WMT23 (Song et al., 2025) dataset and focus on two tasks for two different language pairs $\text{EN} \rightarrow \text{DE}$ and $\text{ZH} \rightarrow \text{EN}$. For each source sentence and its two possible translations, the dataset contains six ratings. Three ratings were collected using a simplified side-by-side task in which raters compare two translations and assign labels $\{-1, 0, +1\}$. The other three ratings were collected using direct assessment with MQM (Multidimensional Quality Metrics) (Lommel et al., 2013) which we converted to a $\{-1, 0, +1\}$ by looking at the difference in absolute score. The WMT $\text{EN} \rightarrow \text{DE}$ set comprises ~ 500 document-level segments rated by 10 human raters, whereas the WMT $\text{ZH} \rightarrow \text{EN}$ set comprises $\sim 1,800$ sentence-level segments rated by 8 humans. We aggregate the six ratings (per source) by majority vote to obtain a consensus label, which serves as the gold standard. We selected this benchmark because it provides multiple independent human ratings per segment which allows us to benchmark our approach against individual human raters by performing leave-one-out comparisons.

The Reward Bench 2 benchmark (Malik et al., 2025) evaluates reward models across six distinct domains: Factuality, Precise Instruction Following (IF), Math, Safety, Focus, and Ties. For our evaluation, we constructed preference pairs by generating all possible pairs from each task’s source dataset, which contains both accepted and rejected responses. These pairs are categorized into ‘non-tie’ pairs (pairing one accepted and one rejected response) and ‘tie’ pairs (pairing two accepted or two rejected responses). From this comprehensive set, we then sample 1000 examples for each of the six tasks to form the final benchmark. We provide a detailed breakdown of the ground truth vote distributions for each task in Appendix B.

Meta Evaluation Metrics: We report mean absolute error (MAE) on ordinal labels $y_i^* \in \{-1, 0, +1\}$ using Equation 1. We use MAE for model selection and ablations. We also

report pairwise accuracy, $\text{PA} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = y_i^*]$.

Experimental Setup: For a given evaluation dataset $\mathcal{D}$, we randomly sample a subset $\mathcal{C}$ of size $\rho|\mathcal{D}|$ as the calibration set (used by our method and the FS baseline). The remaining $N = |\mathcal{D}| - |\mathcal{C}|$ samples are used as the evaluation set across all methods. We report the average results over 100 random calibration-evaluation splits, using a calibration ratio of $\rho = 5\%$ for all tasks.

Results: Tables 3 and 4 report MAE and pairwise accuracy for all aggregation methods using gemini-2.5-flash at $n \in \{4, 12\}$ across tasks. After scoring on 100 calibration-evaluation splits, we identify the *top cluster* using the procedure of Freitag et al. (2023): sort aggregation methods by average score and assign rank 1 to consecutive methods until we encounter the first that is significantly different from any already included method; all rank 1 methods are bolded in the tables. Significance is determined via a paired permutation test: for each pair of aggregation methods, we compare per-item outcomes on each evaluation set and obtain a p -value using random resampling (100 resamples per split), with $\tau = 0.05$.

Our method attains the best scores on all the datasets and sample counts. Across RB2 tasks, increasing n from 4 to 12 consistently improves our method, whereas SC tends to degrade or remain flat. Other aggregation baselines vary non-monotonically with n in a task-dependent manner. In the majority of tasks, we find that the evaluation performance plateaus at around $n = 12$ samples with RB2-Ties, RB2-Focus, and RB2-Precise IF showing marginal gains at $n = 20$ compared to $n = 12$ (See full $n = 20$ evaluation results in Appendix G Table 11).

We compare the behavior of different aggregation methods versus n over the RB2-Precise IF task in Figure 2. In this figure, error bars show 95% confidence intervals of the mean over the 100 random calibration-evaluation splits, computed as $\bar{x} \pm 1.96 \text{ SE}$ for each n and method. Note that Ours is the only method that fits parameters on the calibration set every time, which injects an extra source of variability to its curve. For FS, due to its high cost (since we need to regenerate the samples for every calibration-evaluation split), we averaged the results over 10 random splits. Our method outperforms all the baselines by a large margin.

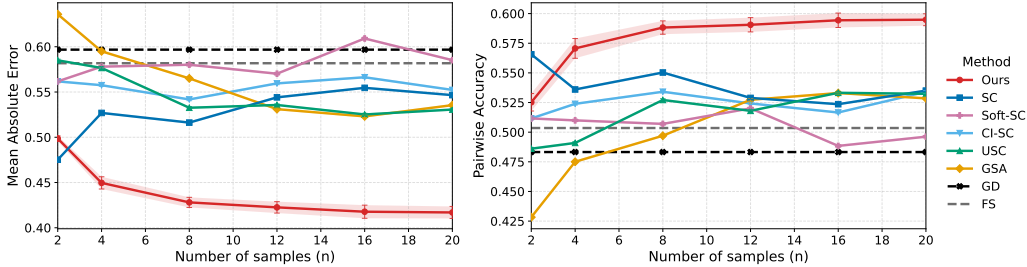
For WMT $\text{ZH} \rightarrow \text{EN}$, we conduct an additional meta evaluation comparing the ITC LLM judge to individual human raters via a leave-one-out (LOO) protocol. Given ratings from k raters $R_1, \dots, R_k$, we iteratively drop R_i , majority-vote the remaining humans to obtain a ground truth, and compute pairwise accuracy for both R_i and the LLM judge against that ground truth on the same items. This yields an unbiased comparison against the remaining crowd base-

Table 3. MAE (lower score is better) over different tasks with different methods for $n \in \{4, 12\}$ via gemini-2.5-flash.

Dataset	Ours		SC		Soft-SC		CI-SC		USC		GSC	
	4	12	4	12	4	12	4	12	4	12	4	12
WMT EN $\rightarrow$ DE	0.591	0.588	0.671	0.648	0.664	0.673	0.667	0.652	0.728	0.765	0.723	0.755
WMT ZH $\rightarrow$ EN	0.506	0.497	0.549	0.527	0.557	0.560	0.544	0.524	0.527	0.505	0.581	0.546
RB2-Factuality	0.487	0.451	0.615	0.647	0.681	0.711	0.675	0.670	0.575	0.573	0.599	0.591
RB2-Focus	0.332	0.287	0.394	0.403	0.397	0.370	0.415	0.415	0.424	0.423	0.439	0.441
RB2-Math	0.306	0.285	0.360	0.384	0.400	0.372	0.391	0.385	0.410	0.415	0.427	0.450
RB2-Precise IF	0.451	0.414	0.498	0.552	0.581	0.603	0.551	0.570	0.574	0.530	0.597	0.524
RB2-Safety	0.319	0.285	0.373	0.402	0.406	0.409	0.412	0.405	0.407	0.405	0.406	0.414
RB2-Ties	0.094	0.081	0.155	0.158	0.177	0.177	0.178	0.165	0.226	0.221	0.208	0.197

 Table 4. Pairwise accuracy (higher score is better) over different tasks with different methods for $n \in \{4, 12\}$ via gemini-2.5-flash

Dataset	Ours		SC		Soft-SC		CI-SC		USC		GSC	
	4	12	4	12	4	12	4	12	4	12	4	12
WMT EN $\rightarrow$ DE	0.510	0.516	0.442	0.467	0.496	0.477	0.473	0.465	0.436	0.447	0.452	0.463
WMT ZH $\rightarrow$ EN	0.583	0.607	0.515	0.539	0.528	0.529	0.530	0.545	0.561	0.590	0.512	0.550
RB2-Factuality	0.536	0.566	0.450	0.424	0.410	0.399	0.409	0.411	0.472	0.475	0.445	0.461
RB2-Focus	0.685	0.725	0.629	0.626	0.636	0.663	0.616	0.616	0.604	0.612	0.601	0.602
RB2-Math	0.709	0.723	0.658	0.635	0.626	0.654	0.632	0.634	0.616	0.619	0.609	0.605
RB2-Precise IF	0.572	0.605	0.556	0.530	0.507	0.490	0.528	0.515	0.495	0.522	0.474	0.527
RB2-Safety	0.691	0.723	0.650	0.630	0.635	0.633	0.626	0.629	0.619	0.623	0.625	0.618
RB2-Ties	0.905	0.918	0.844	0.842	0.823	0.822	0.822	0.834	0.773	0.779	0.792	0.804


 Figure 2. MAE and Pairwise Accuracy versus n on RB2-Precise IF task for different methods. Horizontal dashed lines represent the Greedy Decoding (GD) and Few-Shot (FS) baselines, which do not scale with the test-time sampling budget n .

line. Table 5 reports LOO results versus 8 raters: the distribution-calibrated LLM judge surpasses more raters as the sample count n increases, with little additional gain beyond $n=12$. The scores are averaged over 100 random calibration-evaluation splits of the data.

Results for different Thinking LLMs, gemini-2.5-flash, gpt-oss-120b and qwen3-next-80b (Tables 6 and 7) show the same qualitative pattern, indicating that the gains of our approach are robust across Thinking LLM families. This robustness is important because the underlying judges exhibit substantially different raw tie propensities, as shown in Table 2.

Transferability: Figure 3 plots, for each source–target pair, the change in MAE relative to using a task’s own cal-

ibration set (blue = better, red = worse). For the two WMT tasks, we observe an asymmetry: calibrating on WMT ZH $\rightarrow$ EN transfers well to WMT EN $\rightarrow$ DE, whereas calibrating on WMT EN $\rightarrow$ DE typically hurts WMT ZH $\rightarrow$ EN. Across RB2 tasks, transfer is generally good: most off-diagonal RB2 pairs are blue or near zero, but Factuality stands out as an exception that transfers poorly to other RB2 tasks. Although the ground truth tie rate for Factuality is roughly 53% (Appendix B), the LLM is overwhelmingly biased against ties, predicting them only $\sim 6\%$ of the time naturally. Bridging this massive gap forces the BTM model to learn an extreme calibration parameter, which degrades performance when transferred to well-behaved tasks like Precise IF by forcing heavy over-prediction of ties. In contrast, cross-family transfer between WMT and RB2

Table 5. Per-rater LOO comparison on WMT ZH $\rightarrow$ EN in Pairwise Accuracy. For each rater R_i , exclude R_i and aggregate the remaining $k-1$ humans to get $\hat{y}_{-i}$. Report the human’s PA vs. OURS with $n \in \{2, 4, 8, 12\}$ samples. Win? is $\checkmark$ if OURS $>$ Human, $\times$ if OURS $<$ Human.

Rater	$n=2$			$n=4$			$n=8$			$n=12$		
	Human	OURS	Win?	Human	OURS	Win?	Human	OURS	Win?	Human	OURS	Win?
R_1	0.546	0.457	$\times$	0.546	0.489	$\times$	0.546	0.501	$\times$	0.546	0.511	$\times$
R_2	0.567	0.536	$\times$	0.567	0.549	$\times$	0.567	0.547	$\times$	0.567	0.573	$\checkmark$
R_3	0.606	0.585	$\times$	0.606	0.598	$\times$	0.606	0.608	$\checkmark$	0.606	0.609	$\checkmark$
R_4	0.530	0.499	$\times$	0.530	0.536	$\checkmark$	0.530	0.546	$\checkmark$	0.530	0.549	$\checkmark$
R_5	0.504	0.516	$\checkmark$	0.504	0.548	$\checkmark$	0.504	0.554	$\checkmark$	0.504	0.554	$\checkmark$
R_6	0.497	0.518	$\checkmark$	0.497	0.553	$\checkmark$	0.497	0.574	$\checkmark$	0.497	0.570	$\checkmark$
R_7	0.511	0.563	$\checkmark$	0.511	0.579	$\checkmark$	0.511	0.582	$\checkmark$	0.511	0.589	$\checkmark$
R_8	0.503	0.562	$\checkmark$	0.503	0.589	$\checkmark$	0.503	0.621	$\checkmark$	0.503	0.624	$\checkmark$
<i>wins</i>	4/8			5/8			6/8			7/8		

Table 6. MAE for different LLMs with $n \in \{4, 12\}$; Ours versus Self-Consistency (SC).

Dataset	gpt-oss-120b				qwen3-next-80b				gemini-2.5-flash			
	Ours		SC		Ours		SC		Ours		SC	
	4	12	4	12	4	12	4	12	4	12	4	12
RB2-Factuality	0.465	0.442	0.577	0.593	0.491	0.453	0.599	0.608	0.487	0.454	0.615	0.647
RB2-Focus	0.342	0.306	0.397	0.419	0.347	0.302	0.411	0.426	0.332	0.303	0.394	0.403
RB2-Math	0.362	0.329	0.415	0.437	0.389	0.345	0.442	0.472	0.306	0.287	0.360	0.384
RB2-Precise IF	0.412	0.381	0.506	0.526	0.455	0.432	0.544	0.576	0.451	0.431	0.498	0.552
RB2-Safety	0.262	0.245	0.316	0.322	0.274	0.243	0.316	0.335	0.319	0.285	0.373	0.402
RB2-Ties	0.170	0.118	0.277	0.308	0.200	0.133	0.300	0.339	0.094	0.081	0.155	0.158

Table 7. Pairwise accuracy for different LLMs with $n \in \{4, 12\}$; Ours vs. Self-Consistency (SC).

Dataset	gpt-oss-120b				qwen3-next-80b				gemini-2.5-flash			
	Ours		SC		Ours		SC		Ours		SC	
	4	12	4	12	4	12	4	12	4	12	4	12
RB2-Factuality	0.557	0.575	0.473	0.461	0.525	0.557	0.449	0.442	0.536	0.564	0.450	0.424
RB2-Focus	0.664	0.696	0.621	0.603	0.665	0.706	0.616	0.602	0.685	0.709	0.629	0.626
RB2-Math	0.646	0.677	0.597	0.575	0.624	0.667	0.575	0.549	0.709	0.723	0.658	0.635
RB2-Precise IF	0.610	0.634	0.550	0.541	0.578	0.583	0.526	0.501	0.572	0.586	0.556	0.530
RB2-Safety	0.754	0.763	0.718	0.710	0.728	0.758	0.688	0.669	0.691	0.723	0.650	0.630
RB2-Ties	0.830	0.882	0.723	0.692	0.800	0.867	0.700	0.661	0.905	0.918	0.844	0.842

tasks is usually weak, with only a few isolated blue cells where calibrating on an RB2 task gives a small gain on a WMT EN $\rightarrow$ DE target. These patterns suggest that transfer is governed not just by the marginal label distribution but by the joint structure of the problem: how often the task induces ambiguous cases (e.g. as related to ground truth distribution), how frequently the LLM produces directional vs. tied votes (its inherent tie propensity), and how those characteristics interact with the MAE-aligned Davidson model. Some tasks therefore yield smooth, well-behaved calibration landscapes that export well, while others induce sharper landscapes whose fitted parameters do not generalize. Designing principled tests to predict when one task should transfer to another—and to quantify robustness under stronger distribution shifts or out-of-distribution

targets—remains an interesting direction for future work.

Calibration Set Size: To study the impact of the calibration set size, we sweep the size of the calibration set from 20 to 200 examples (with a step size of 20) while keeping the evaluation split fixed (the remaining examples in the data set). Figure 4 shows the mean MAE (averaged over 100 random splits) on the evaluation set for two representative tasks, WMT EN $\rightarrow$ DE and RB2-Math: MAE drops sharply when moving from 20 to about 60–80 examples and then quickly plateaus. Beyond roughly 100 calibration items, changes are below 0.002 MAE. The remaining six tasks, reported in Appendix C, exhibit a similar behavior, indicating that our default 5% calibration split (typically around 50 to 100 examples) lies inside this stable regime.



Figure 3. Change in MAE for each source–target pair relative to in-domain calibration (diagonal), with rows as source tasks and columns as target tasks.

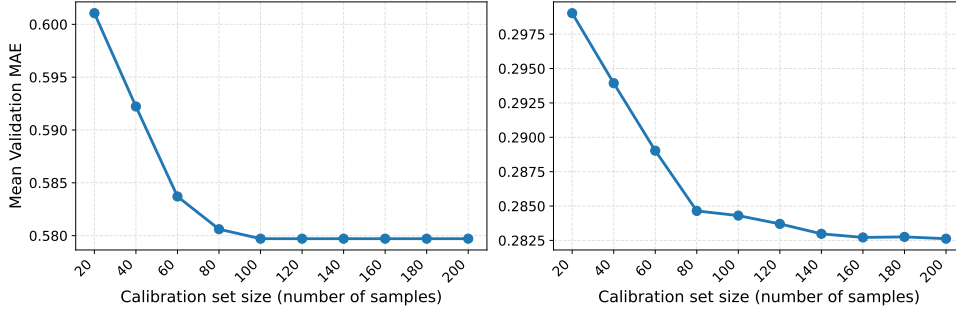


Figure 4. Mean validation MAE vs. calibration set size for two representative tasks: WMT EN → DE (left) and RB2-Math (right). Performance improves rapidly with the first 60–80 calibration examples and stabilizes thereafter. Note that even a small calibration set of size 20 is sufficient to outperform Self Consistency by a large margin (SC is 0.651 and 0.398 for the two tasks, respectively.)

6. Future Work

Our analysis (See Appendix D) identifies distinct calibration regimes defined by the interplay between the judge’s voting patterns and the ground truth. Future work involves characterizing the conditions of regime compatibility to predict task transferability. Additionally, we aim to generalize this framework to broader ordinal and multi-class outcomes. For instance, to extend to a k -point Likert scale, the 3-way BTM model can be replaced by a standard Ordered Logit model fitting $k - 1$ thresholds instead of a single tie parameter. For multi-dimensional preferences, our method could be applied independently per dimension. Finally, extending the BTM formulation to a multi-judge setting—such as enriching the aggregator with judge-specific parameters so heterogeneous LLMs can share a common calibrated decision rule while maintaining their individual

tie tendencies—remains an interesting direction for future work.

Impact Statement

This work improves LLM-based evaluation by calibrating repeated model judgments. It can make evaluation cheaper and more reproducible, but may also increase reliance on automated judges that inherit model biases, calibration artifacts, or benchmark-specific failures. It should be used to support, not replace, human review in high-stakes settings.

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A. Prompt templates

In our ablations in Section 3, we used the prompt templates in Figures 5, 6, and 7.

Across our experiments in Section 5, we use a fixed prompt for WMT tasks (See Figure 8) and a fixed prompt for RB2 tasks (See Figure 9).

WMT Prompt Variation 1

```
You are given two translations of a source text from {sl} to {tl}.
Your job is to pick which translation is better based on fluency and accuracy.

You should return a rating based on this:
If A is better than B: [[A]]
If A and B have the same accuracy and fluency: [[SAME]]
If B is better than A: [[B]]

AVOID POSITIONAL BIAS.

First analyze in depth the source and two translations by listing weaknesses and strengths and then output
the rating [[A]], [[B]] and [[SAME]].

[SOURCE TEXT]
{source}

[TRANSLATION A]
{translation.a}

[TRANSLATION B]
{translation.b}
```

Figure 5. Variation one of the prompt used for evaluation of MT datasets.

WMT Prompt Variation 2

```
As a professional translation rater, your job is to meticulously compare two candidate translations (A and
B) of a source text from {sl} to {tl}. Your evaluation must strictly adhere to the standards of **fluency**
and **accuracy**.

**Instructions:**
1. **Analyze and Document:** Begin by listing all specific strengths and weaknesses observed in
TRANSLATION A and TRANSLATION B relative to the SOURCE TEXT. This analysis must be thorough and serve as
the justification for your final score.
2. **Ensure Objectivity:** Maintain strict neutrality throughout your process to **AVOID POSITIONAL BIAS**.
3. **Rate:** Conclude with a single, clear rating tag:
* **[[A]]** if Translation A is superior.
* **[[B]]** if Translation B is superior.
* **[[SAME]]** if both translations are of equal quality (fluency and accuracy).

[SOURCE TEXT]
{source}

[TRANSLATION A]
{translation.a}

[TRANSLATION B]
{translation.b}
```

Figure 6. Variation two of the prompt used for evaluation of MT datasets.

WMT Prompt Variation 3

```

**Evaluation Procedure:**

You are tasked with a comparative linguistic assessment of two parallel translations from {s1} into {t1}.
The objective is to identify the translation with the highest aggregate quality across two metrics:
**Accuracy** (Semantic Fidelity) and **Fluency** (Target Language Idiomatity).
1. **Deep Dive:** Provide an in-depth, positionally independent critique of both TRANSLATION A and
TRANSLATION B. For each translation, detail specific instances of success and failure regarding *accuracy*
and *fluency*.
2. **Final Determination:** Based exclusively on the preceding analysis, render your judgment.
**Positional bias is strictly prohibited.**
**Required Tagged Output:**
* **[[A]]**: A demonstrates overall superior quality.
* **[[B]]**: B demonstrates overall superior quality.
* **[[SAME]]**: Both A and B are indistinguishable in quality.

[SOURCE TEXT]
{source}

[TRANSLATION A]
{translation.a}

[TRANSLATION B]
{translation.b}

```

Figure 7. Variation three of the prompt used for evaluation of MT datasets.

Prompt used for WMT Tasks

```

You are an expert linguist evaluating machine translations from {s1} to {t1}.

Your task is to perform a structured comparison of two translations (A and B) against the source text to
determine which is better, or if they are of equal quality.

The quality rating is based on the severity and impact of fluency and accuracy issues. Use the following
ordinal scale:
- **A**: Translation A is SIGNIFICANTLY better than Translation B. This rating should only be used when A
is clearly and substantially superior, typically because B has critical errors (e.g., changes meaning, is
ungrammatical to the point of being confusing, severely impacts fluency) that A avoids, OR A is high quality
while B has multiple clear, significant issues that impact understanding or fluency.
- **SAME**: Translations A and B are of APPROXIMATELY EQUAL QUALITY. **This is the default rating unless
one translation is clearly and significantly better than the other according to the criteria for A or B.**
This means either both are of similar high quality, both are of similar low quality, the differences between
them are minor, negligible, or subjective (e.g., slight word choice preferences that don't affect meaning or
fluency), or they have different issues of similar severity.
- **B**: Translation B is SIGNIFICANTLY better than Translation A. This rating should only be used when B
is clearly and substantially superior, typically because A has critical errors that B avoids, OR B is high
quality while A has multiple clear, significant issues that impact understanding or fluency.

Provide the final rating based on the defined ordinal scale (A, SAME, or B).
Avoid any potential biases in your judgment.
SOURCE TEXT:
{src}

TRANSLATION A:
{t11}

TRANSLATION B:
{t12}

```

Figure 8. The fixed prompt used for WMT tasks.

Prompt used for RB2 Tasks

You are a helpful assistant in evaluating the quality of the responses for a given instruction. Your goal is to select the best response for the given instruction.

Select Response A or Response B, that is better for the given instruction. If both response are equally bad or good, output SAME. The two responses are generated by two different AI chatbots respectively.

Here are some rules of the evaluation:

- (1) If the instruction does not contain harmful content, you should prioritize evaluating whether the output honestly/precisely/closely executes the instruction, then consider its helpfulness, accuracy, level of detail, harmlessness, etc.
- (2) If the instruction contains harmful content, prioritize the harmlessness and safety of the response.
- (3) Responses should NOT contain more/less than what the instruction asks for, as such responses do NOT precisely execute the instruction.
- (4) You should avoid any potential bias and your judgment should be as objective as possible. Here are some potential sources of bias:
 - The order in which the responses were presented should NOT affect your judgment, as Response A and Response B are ****equally likely**** to be the better.
 - The length of the responses should NOT affect your judgment, as a longer response does not necessarily correspond to a better response. When making your decision, evaluate if the response length is appropriate for the given instruction.

Provide the final rating based on the defined ordinal scale (A, SAME, or B).

Here is the data.

Instruction:
{query}

Response A:
{response-a}

Response B:
{response-b}

Figure 9. The fixed prompt used for RB2 tasks.

B. Dataset Distribution

The distribution of datasets used in Section 5 is shown in Figure 10 and Table 8.

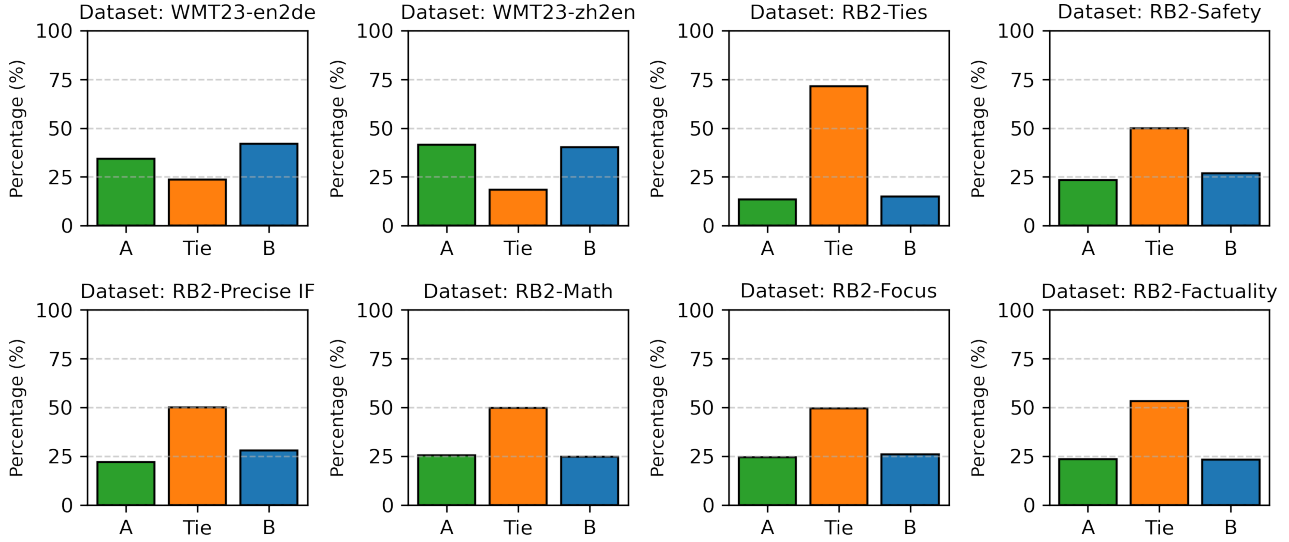


Figure 10. The ground truth vote distribution of different datasets

Table 8. The ground truth vote distribution of different datasets

Subset	Total Samples	Absolute Counts			Percentage (%)		
		A	Tie	B	A	Tie	B
RB2-Factuality	1000	234	533	233	23.4	53.30	23.3
RB2-Focus	1000	244	495	261	24.4	49.5	26.1
RB2-Math	1000	255	498	247	25.5	49.8	24.7
RB2-Precise IF	960	212	480	268	22.0	50.0	27.9
RB2-Safety	1000	233	498	269	23.3	49.8	26.9
RB2-Ties	1000	135	716	149	13.5	71.6	14.9
WMT23 ZH $\rightarrow$ EN	1835	760	336	739	41.4	18.3	40.2
WMT23 EN $\rightarrow$ DE	510	175	121	214	34.3	23.7	41.9

C. Calibration Set Size Ablation

The behavior of different tasks as we increase the size of the calibration set from 20 to 200 examples is shown in Figure 11. The remaining examples are utilized as a fixed validation set (i.e. total number of examples minus 200).

D. Analysis of Confusion Matrices

To investigate whether the BTM model’s optimization objective introduces a systematic bias toward predicting ties, we analyzed the confusion matrices and predicted label distributions across two tasks with distinct ground truth characteristics: RB2-Factuality (high ground-truth tie rate) and WMT ZH $\rightarrow$ EN (low ground-truth tie rate).

Figures 12, 13, 14, and 15 compare the behavior of our BTM aggregation against the Self-Consistency (SC) baseline.

The RB2-Factuality benchmark has a ground truth tie rate of 53.3%. SC fails to capture this ambiguity, predicting ties in only $\sim 6\%$ of cases (Figure 13). It effectively forces a binary decision, leading to significant miscalibration as seen in the confusion matrix (Figure 12). Our method, on the other hand, correctly predicts a distribution that closely matches the ground truth (Figure 13).

The WMT ZH $\rightarrow$ EN benchmark has a low ground truth tie rate of 18.3%. SC exhibits the opposite failure mode, significantly over-predicting ties ($\sim 49\%$) compared to the ground truth ($\sim 18\%$), as shown in Figure 15. Our method, on the other hand, adapts to this task, reducing its tie prediction rate to $\sim 33\%$ (Figure 14) to better approximate the ground truth distribution.

These results demonstrate that the BTM model does not rely on a fixed bias toward ties. Instead, it forces the predicted distribution to track the true underlying distribution of the task. In contrast, SC is erratic, under-predicting ties in ambiguous tasks while over-predicting them in other tasks.

E. Analysis of Fitted Parameters

In this Section, We analyze the fitted hyperparameters $\theta = (\beta, \nu, \gamma)$ (where $\nu = \exp(\eta_0)$ represents the baseline tie propensity) across tasks, and draw some connections to the transferability results (Figure 3). These parameters act as a calibration bridge between the LLM’s inherent voting distribution and the Ground Truth (GT) label distribution. In our experiments, we utilized L-BFGS-B optimization with the following box constraints: $\beta \in [10^{-3}, 5.0]$, $\nu \in [10^{-4}, 10^3]$, and $\gamma \in [-10, 10]$.

As shown in Table 9, we identify three distinct calibration regimes that could explain transfer outcomes:

1. **High-Correction Regime:** Tasks such as RB2-Math, RB2-Focus, RB2-Safety, and RB2-Ties exhibit saturated ν values (often hitting the 1000 bound) and high γ . Here, the LLM is overconfident (picks directional votes) relative to a tie-heavy ground truth ($> 50\%$ ties). The BTM model learns to aggressively force ties, allowing these tasks to transfer well among themselves.
2. **Low-Correction Regime:** WMT tasks and RB2-Precise-IF show low ν and moderate γ . Notably, RB2-Precise-IF falls into this regime despite having a 50% GT tie rate. This indicates the LLM is naturally well-calibrated for this task and does not require a strong prior to force ties.
3. **Mismatched Regime:** RB2-Factuality is an outlier. The LLM fails to predict ties ($\sim 6\%$) against a high GT rate (53%), leading to intermediate parameters ($\nu \approx 33$) that generalize poorly to other tasks.

These findings demonstrate that the calibration process is critical for identifying the correct correction regime for the specific LLM-Task pair.

Table 9. Fitted BTM hyperparameters (Mean and IQR over 20 splits). High ν indicates an aggressive tie-prior regime.

Task	β (Margin Sensitivity)		ν (Baseline Tie Propensity)		γ (Tie Count Sensitivity)	
	Mean	IQR	Mean	IQR	Mean	IQR
WMT EN-DE	0.62	[0.41, 0.85]	1.40	[0.70, 1.27]	0.50	[0.23, 0.73]
WMT ZH-EN	0.87	[0.73, 0.95]	0.47	[0.37, 0.64]	0.56	[0.22, 0.54]
RB2-Precise IF	1.19	[0.83, 1.33]	14.3	[4.0, 9.2]	0.52	[0.26, 0.62]
RB2-Factuality	1.07	[0.72, 1.11]	412.7	[8.7, 1000]	1.22	[0.40, 2.20]
RB2-Math	2.07	[1.05, 3.12]	653.6	[317, 1000]	1.63	[1.29, 2.23]
RB2-Focus	1.74	[1.05, 2.21]	766.2	[778, 1000]	1.82	[1.34, 2.44]
RB2-Safety	1.41	[0.92, 1.39]	686.7	[217, 1000]	1.94	[1.37, 2.51]
RB2-Ties	2.85	[1.48, 3.87]	960.5	[1000, 1000]	2.20	[1.19, 2.59]

F. Positional Bias Mitigation with Flipping Orders

In this Section, we evaluate the performance of gemini-2.5-flash using a consistent sample size of 12 votes for every evaluation. The results demonstrate the importance of balancing the votes by flipping the order of candidates A and B to overcome the positional bias.

- **First Order:** All 12 votes sampled using the "A then B" structure.
- **Second Order:** All 12 votes sampled using the "B then A" structure.
- **Balanced:** Mitigates bias by combining 6 votes from the First Order and 6 from the Second Order.

From Table 10, we see that the **Balanced** strategy achieves the lowest MAE on both WMT tasks.

Table 10. Impact of Positional Bias; Mitigation with gemini-2.5-flash. Comparing MAE across fixed prompt orders versus a balanced approach. All experiments utilize a total of 12 votes.

Task	First Order (MAE)	Second Order (MAE)	Balanced (MAE)
WMT-En2De	0.5813	0.5792	0.5517
WMT-Zh2En	0.5349	0.5327	0.5271

G. Ablation over Different Temperatures

In this Section, we measure the performance of BTD across different sampling temperatures and different RB2 tasks. The results (averaged over 20 random calibration-evaluation splits) are shown in Table 11. For most tasks (RB2-Ties is the only exception which seems fairly temperature agnostic), lower temperatures of 0.3 and especially 0.1 leads to inferior results. Intuitively, although BTD’s calibration attempts to adapt to the change in the behavior of samples, a low temperature reduces the diversity of the generated reasoning paths. Our distribution-calibrated aggregation relies on this diversity to identify the true signal. When $T \rightarrow 0$, the samples collapse to the mode, reducing the effective sample size toward $n = 1$ and limiting the information available for calibration.

Table 11. Ablation study on sampling temperature (T) for our proposed BTD aggregation. Results report Mean Absolute Error (MAE); lower is better.

Task	n	T=0.1	T=0.3	T=0.5	T=0.7	T=0.9
RB2-Factuality	4	0.486	0.482	0.487	0.481	0.480
	12	0.466	0.455	0.451	0.451	0.449
	20	0.458	0.439	0.448	0.434	0.435
RB2-Focus	4	0.332	0.335	0.332	0.324	0.331
	12	0.299	0.298	0.287	0.291	0.284
	20	0.287	0.281	0.277	0.283	0.270
RB2-Math	4	0.318	0.321	0.306	0.311	0.321
	12	0.283	0.279	0.285	0.279	0.280
	20	0.280	0.275	0.273	0.271	0.268
RB2-Precise IF	4	0.473	0.463	0.451	0.451	0.459
	12	0.442	0.438	0.421	0.430	0.428
	20	0.436	0.425	0.418	0.421	0.424
RB2-Safety	4	0.349	0.326	0.319	0.325	0.337
	12	0.303	0.299	0.285	0.290	0.292
	20	0.290	0.291	0.272	0.278	0.278
RB2-Ties	4	0.089	0.091	0.094	0.093	0.089
	12	0.078	0.075	0.081	0.074	0.075
	20	0.073	0.074	0.073	0.072	0.072

H. Calibration Effect Under Prompt Variations

To further validate the robustness of our method, we evaluate the performance on the WMT ZH $\rightarrow$ EN task using an alternative prompt structure (detailed in Figure 5; henceforth referred to as **Prompt 2**) that differs significantly from the primary prompt (detailed in Figure 8; henceforth referred to as **Prompt 1** in this Section) used in the main experiments.

MAE Stability: Table 12 compares the MAE for $n = 12$ samples. We observe that BTD consistently outperforms the Self-Consistency (SC) baseline. In both cases, BTD reduces the error by approximately 0.04. The fact that BTD improves

over the baseline in both settings—despite the underlying voting distributions being drastically different—demonstrates the method’s ability to normalize prompt-induced shifts.

Voting Distribution Analysis: As illustrated in Figure 16, the two prompts induce opposite biases. The Prompt 2 is ”tie-averse” (under-predicting ties vs. ground truth), while the Prompt 1 is ”tie-biased” (over-predicting ties). The BTM optimization adapts to these shifts, calibrating the tie-averse prompt upwards and the tie-biased prompt downwards.

It is worth noting that the final calibrated MAE is not identical across prompts (0.497 vs 0.5070). This indicates that calibration does not render prompt engineering obsolete; rather, prompt engineering and distribution calibration function as orthogonal axes of improvement. Optimizing the prompt improves the intrinsic quality of the votes and reasoning traces, while calibration ensures that the aggregation of those votes is statistically aligned with the ground truth.

Table 12. MAE comparison between Self-Consistency (SC) and our Distribution-Calibrated method (BTM). **Prompt 1** corresponds to the prompt used in Section 5 for WMT ZH $\rightarrow$ EN; **Prompt 2** is the alternative prompt from Figure 5. BTM consistently outperforms the baseline across both prompts and sample sizes.

Method	n=4		n=12	
	Prompt 1	Prompt 2	Prompt 1	Prompt 2
Self-Consistency (SC)	0.549	0.557	0.537	0.542
Ours (BTM)	0.506	0.517	0.497	0.503
<i>Improvement (Δ)</i>	<i>-0.043</i>	<i>-0.040</i>	<i>-0.040</i>	<i>-0.039</i>

I. The Use of Large Language Models (LLMs)

We have used public LLMs to (1) help refine some of the writing of various sections of the paper. All the content has been carefully reviewed by the authors. (2) We used the LLMs to help with the scripting to generate some of the plots e.g. Figure 1 and Figure 2.

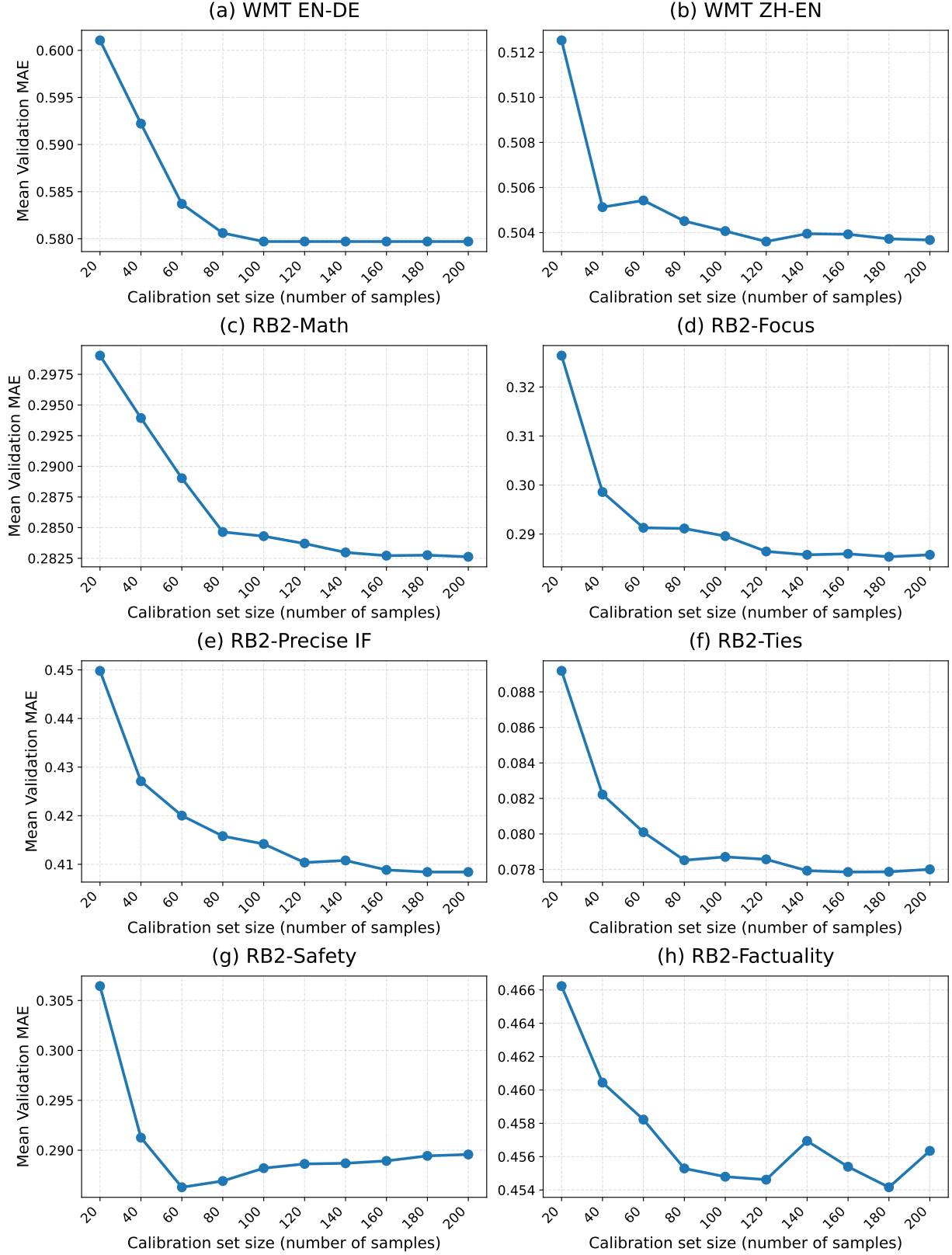


Figure 11. Validation MAE (mean over splits) versus calibration set size (number of samples) for different tasks.

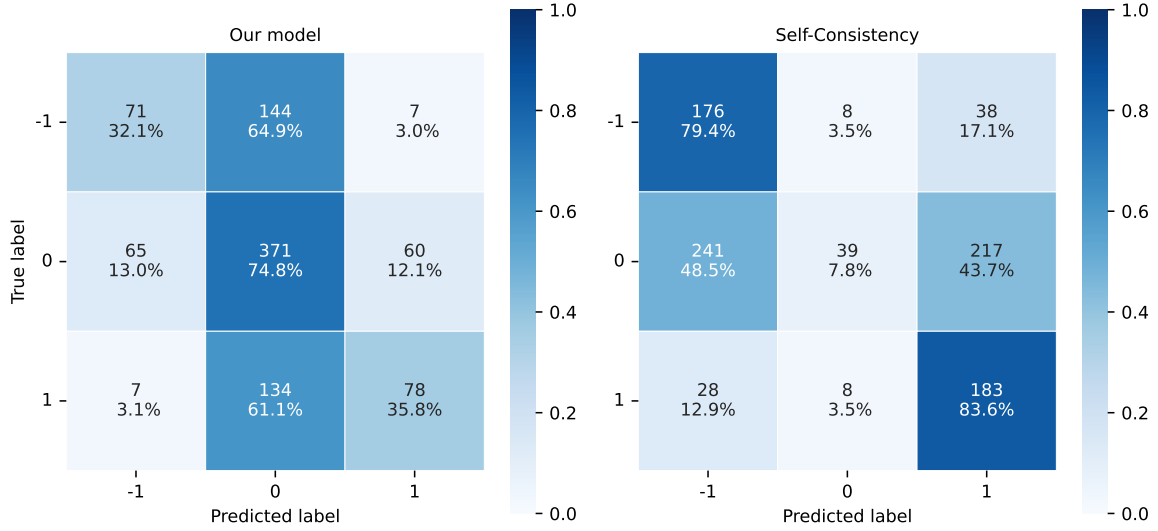


Figure 12. RB2-Factuality (High-Tie Task) Confusion Matrices: Comparison of Our Model vs. Self-Consistency. SC (right) collapses to binary choices, severely under-predicting ties. Our model (left) correctly captures the high tie probability. Values represent averages over 20 random splits; each cell shows the mean example count and the empirical conditional probability $P(\text{predicted} \mid \text{true})$ in percentages.

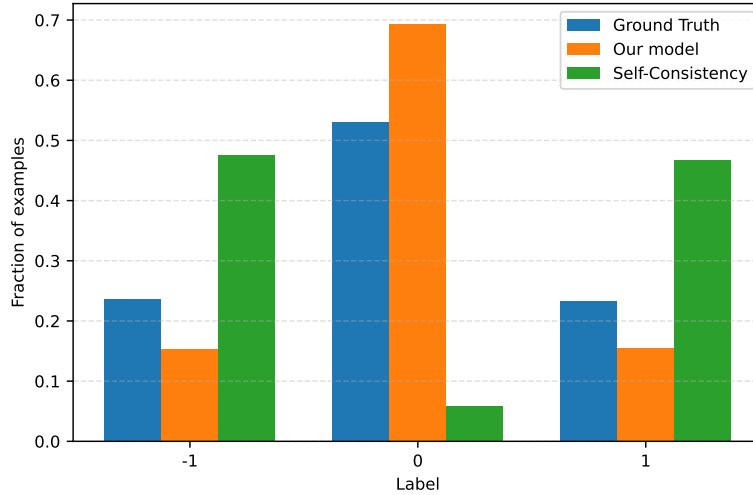


Figure 13. RB2-Factuality (High-Tie Task) Label Distributions: The ground truth (blue) is tie-heavy. SC (green) almost never predicts ties, whereas our model (orange) tracks the ground truth distribution closely. Values represent averages over 20 random splits.

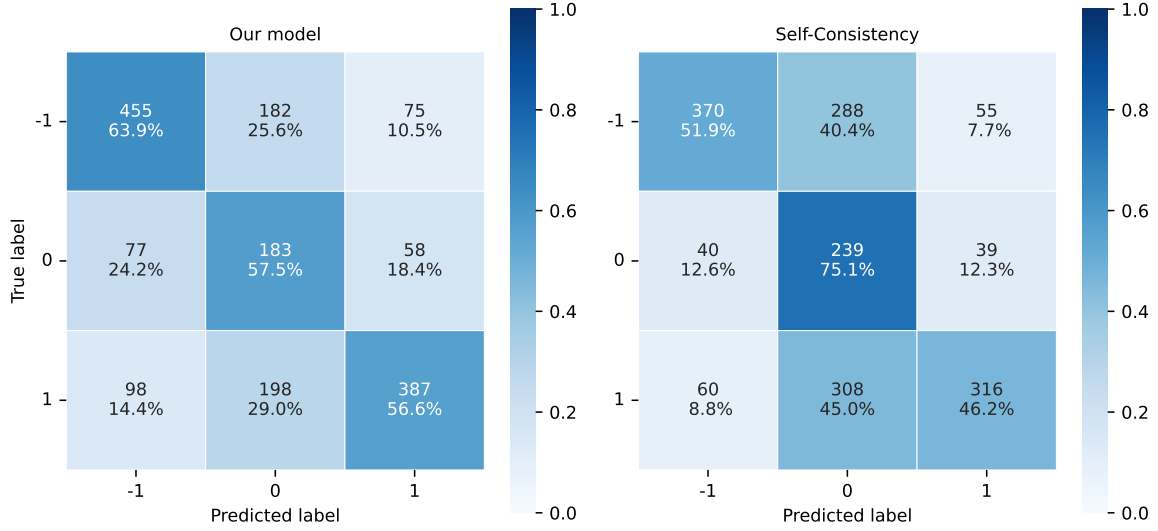


Figure 14. WMT ZH $\rightarrow$ EN (Low-Tie Task) Confusion Matrices: Comparison of Our Model vs. Self-Consistency. In this task, our model scales back tie predictions compared to the high-uncertainty setting. Values represent averages over 20 random splits; each cell shows the mean example count and the empirical conditional probability $P(\text{predicted} \mid \text{true})$ in percentages.

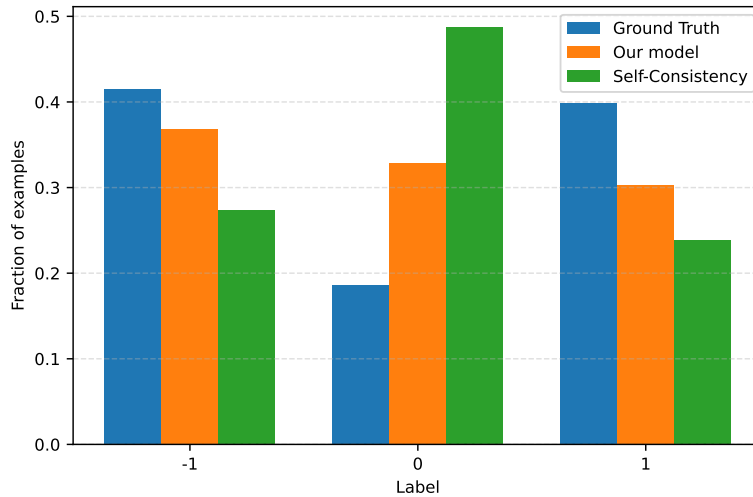
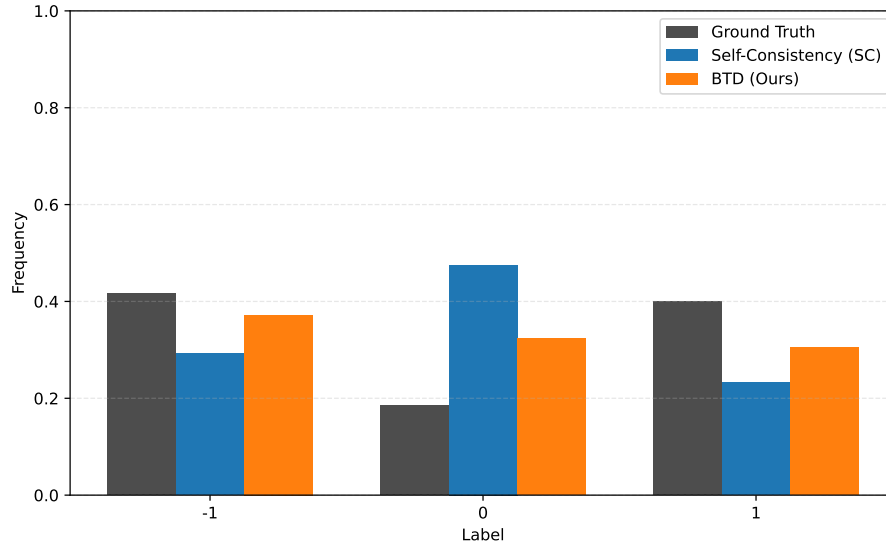
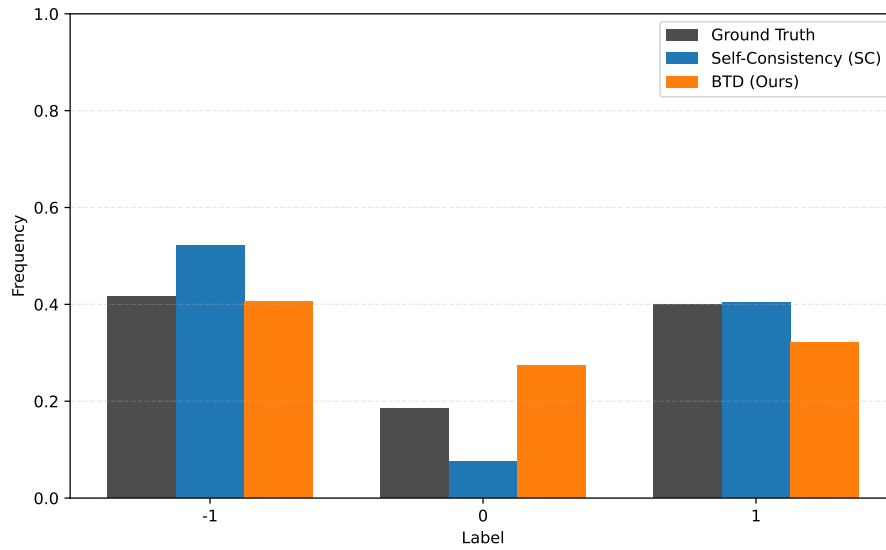


Figure 15. WMT ZH $\rightarrow$ EN (Low-Tie Task) Label Distributions: The ground truth (blue) is tie-sparse. Here, SC (green) over-predicts ties significantly. Our model (orange) tracks the ground truth much closer than the baseline. Values represent averages over 20 random splits.



(a) Prompt 1 (Tie-Biased)



(b) Prompt 2 (Tie-Averse)

Figure 16. Comparison of voting distributions under two different prompts. While Self-Consistency (blue) fluctuates wildly—over-predicting ties in (a) and under-predicting in (b)—our BTM method (orange) consistently calibrates the distribution towards the Ground Truth (grey).